Table 14.2 Distribution of Antigen (RBC) and Antibody (Plasma) in different Blood Groups

Blood Group	Antigens on RBC	Antibodies in Plasma	Can donate to	Can receive from
A	Antigen A	anti- b	A and AB	A and O
B	Antigen B	anti- a	B and AB	B and O
AB	Antigen A and B	No antibody	AB	A, B, AB and O (Universal Recipient)
O	No Antigen	Both anti a and b	A, B, AB and O (Universal Donor)	O

persons do not have Rh antigen on the surface of RBC. Antibodies developed against this Rh antigen is called **Rh antibodies.**

14.18 Lymphatic System

The lymphatic system comprises of lymphatic capillaries, lymphatic vessels, lymph nodes and lymphatic ducts. Lymph is the fluid that flows through the lymphatic system.

The **lymphatic capillaries** unite to form large **lymphatic vessels**. **Lymph nodes** are small oval or pear shaped structures located along the length of lymphatic vessels.

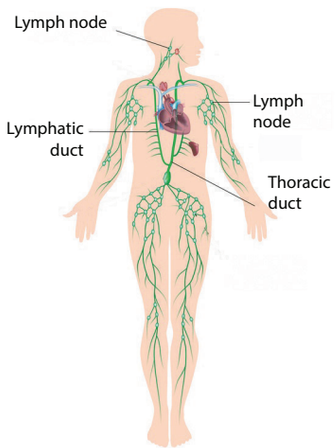


Figure 14.16 Lymphatic System in Man

Lymph

Lymph from the intercellular spaces drains into lymphatic capillaries. Lymph is a colourless fluid formed when plasma, proteins and blood cells escape into intercellular spaces in the tissues through the pores present in the walls of capillaries. It is similar to blood plasma, but is colourless and contains less proteins.The lymph contains very small amount of nutrients, oxygen, CO_2, water and WBC.

Functions of Lymph

- Supplies nutrients and oxygen to those parts where blood cannot reach
- It drains away excess tissue fluid and metabolites and returns proteins to the blood from tissue spaces.
- The lymph also carries absorbed fats from small intestine to the blood. The lymphatic capillaries of intestinal villi (lacteals) absorb digested fats.
- Lymphocytes in the lymph defend the body from infections.

Points to Remember

- The movement of molecules from a region of higher concentration to a region of their lower concentration without the utilization of energy is called diffusion.
- Osmosis is the movement of solvent or water molecules from the region of higher concentration to the region of lower concentration through a semi-permeable membrane.
- Transpiration is the evaporation of water in plants through stomata in the leaves.
- The circulatory system consists of the circulating fluids, the blood and lymph and the heart and its blood vessels.

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